IMPLEMENTATION OF ON-LINE ZONE CONTROL STRATEGIES FOR OPTIMAL RAMP METERING IN THE MINNEAPOLIS RING ROAD

Y J Stephanedes

University of Minnesota, USA

INTRODUCTION

Over the past decade, U.S. traffic operations have steadily evolved toward the concept of network-wide freeway ramp control. Abandoning the practice of isolated control of individual ramps, an increasing number of operations have adopted zonal control, in which sets of ramps along a freeway zone or section are controlled in a coordinated manner. Zonal control has been found more effective in improving the performance of long sections of freeways, where the number of controlled ramps has been rapidly increasing.

Freeway control systems can be classified as open-loop or closed-loop. In open-loop, or pretimed, systems freeway performance has no effect upon the control action at the entrance ramp. Because such systems cannot control freeway performance in the presence of traffic fluctuations, open-loop systems have been decreasing in number in the U.S. since the 1970's.

In closed-loop freeway control systems, measurements of traffic flow, e.g., volume and occupancy, directly affect the ramp control action. The actuating error signal, which is the difference between the desired freeway condition (e.g., desired occupancy) and the measurement, is fed back to the controller so as to bring the performance of the system (e.g., occupancy) to a desired value. The actuating error can be modified by a feedforward signal of upstream freeway conditions, e.g., approaching platoons. Owing to their ability to reduce the sensitivity of traffic control operations to external disturbances, closed-loop freeway control systems, employing both feedback and feedforward control, have been increasing in number in the U.S. over the last two decades. Closed-loop systems are particularly appropriate in operations that have adopted zonal control.

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Closed-loop freeway control systems can be further classified by their ability to modify their action in response to changing system conditions, such as congestion, incidents, weather, and special events. Although the effects of small changes are attenuated by feedback, only an ability to adapt will allow the traffic control to reach the desired traffic performance when such changes are substantial. The first class, on-line calculation systems, are not adaptive; they calculate metering rates in real time based on simple formulas and constant parameters. The second class, gain-scheduling systems, have been increasingly popular in the U.S. over the last 15 years. Gain-scheduling ramp control comes closer to being adaptive as it allows its feedback/feedforward gains to adjust to changing traffic conditions by feedforward compensation. However, there is no feedback from the performance of the closed-loop freeway system that would compensate for an incorrect schedule. Because the parameters, i.e. ramp rates, are changed in an open loop, gain scheduling is not truly adaptive.

Hierarchical control strategies have been proposed for optimum control of freeways and freeway corridors (1-4), and optimization methods have been tested in conjunction with learning mechanisms (5), but these tools have not yet found their way into practice.

This paper presents the results from field implementation of gain scheduling in a feedforward/feedback zonal control system. An optimization algorithm determines the zone-optimal gains based on feedback from the performance of the closed-loop freeway system. Because the parameters, i.e., ramp rates adapt to traffic conditions in a closed loop, the new system is one step closer to being adaptive. The system is a practical alternative to advanced theoretical concepts of freeway corridor control, and can be integrated with gain scheduling currently available at traffic management centers.

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FORMULATION

Two are the major results from this work. First, a control-emulation method that can evaluate traffic responsive ramp-metering strategies to improve freeway performance. Second, an on-line zone-based method for optimal ramp control in instrumented freeway networks.

Control-Emulation

The control-emulation method (6) emulates real time metering, as currently practiced in the U.S. and Canada, and traces the interactions between ramp control strategies and freeway performance through time. The emulation is based on simulated freeway performance when freeway demand patterns and geometrics are known. The ramp control module emulates the on-line rate schedule process typically practiced at Traffic Management Centers or employs optimal metering rates, if available, and is based on traffic information provided by the freeway performance module. The freeway performance module determines traffic state variables, such as volume and occupancy, based on continuum modeling. The method determines the effect of short term, e.g., 30-sec, metering rate changes on traffic performance at the detector stations of the freeway zone.

Optimal Zone Control

The zone control method (6) determines ramp metering rates (gains) based on a gain schedule that is designed with volume/occupancy information received from loop detectors on the main freeway. For instance, in Minneapolis metering rates at a ramp within a zone are determined based on volume and occupancy at up to ten mainline stations upstream (feedforward signal) or downstream (feedback signal) of that ramp. For each ramp, Minnesota Department of Transportation (Mn/DOT) traffic engineers have used historical data to derive a rate schedule of 6 metering rates and a set of corresponding volume-occupancy thresholds. Different schedules can address different traffic and weather conditions.

In this work, a two-step method was developed to optimize the rate schedule in a freeway zone or section. The first step determines the initial schedule for each ramp using the conservation equation and local traffic conditions. The method uses experimental design to determine a set of potential rate schedules. The performance of each schedule is determined by simulation using the control-emulation tool, and the schedule with the best performance is selected. The method could also be applicable in determining initial schedules for a new freeway zone, in which no closed-loop metering control has been implemented.

The second step of the method is initialized with the values determined in the first step, and determines the optimal schedule for the current zone metering control using the control-emulation method and an optimization process. The method determines the optimal schedules for all ramps in a given zone for given traffic demand patterns and management objectives.

METHOD

For determining the initial rate schedule for a ramp entering freeway segment L, the conservation equation provides the rate of change of density (the number of vehicles in L at time t), as a function of input and output volumes to L. Following a transformation, the discrete form of the conservation equation determines future occupancy as a function of current occupancy, and a nonlinear function W(Vu, Vr) of upstream volume Vu and metering rate Vr. Function W is derived from historical data and is updated in real time, completing the closed loop for the determination of the gain schedule and making the system adaptive.

For any desired future occupancy, the corresponding metering rate Vr can be calculated given volume and occupancy values. In practice, however, rate-schedule ramp control systems are discrete, i.e., use a predefined set of metering rates or red times. For example, the Minneapolis and Denver control systems use six rates, Chicago uses five, and the Los Angeles control system uses 15 metering rates.

With the number and values of available red times fixed, the above method can determine the range, i.e. thresholds, of volume and

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occupancy for which each red time should be valid. Each threshold point (Oi, Vi) can be determined from experimental design in conjunction with simulation. The method can determine the threshold values which optimize the desirable performance index, such as total freeway travel time, subject to local traffic conditions.

The best-performing threshold set is used to initialize the optimization process. Optimization, in conjunction with the control-emulation method, determines the threshold set for each schedule, which optimizes the performance of the complete freeway zone or section.

APPLICATION

The new zone-control method was implemented in a field test in Interstate I-494, the ring road south of Minneapolis. The section of westbound I-494 freeway from Portland Ave. to E. Bush Lake Road was selected as a test site in consultation with Mn/DOT Traffic Management Center (TMC). The selected 6.2-mile section includes two zones and contains 19 entrance/exit ramps and a variety of highway geometrics including merging, diverging and weaving areas and the major junction with the I-35W freeway. The junction generates strong weaving effects and congestion that influence the performance of the control at all neighboring ramps. The test section, illustrated in Figure 1, is well instrumented with loop detectors on the main freeway, entrance and exit ramps.

First, a set of metering thresholds was developed for each ramp using the initialization module. For determining the initial thresholds for each ramp, nonlinear function W(Vu, Vr) was parameterized from 5-minute volume and occupancy data over 3 weeks. The parameterized function was also used to determine the desired downstream occupancy. The new thresholds were used to initialize the optimization module that determined, off-line, the optimal thresholds for all ramps by considering systemwide traffic conditions under a given management objective.

Three are the principal objectives of optimizing the ramp metering schedules in the Twin Cities, i.e., reducing freeway delay, keeping freeway traffic running smoothly, and achieving equitable waiting times across entrance ramps. These objectives were achieved using total travel time, which includes waiting time, as the performance index. The optimization method seeks to improve conditions at the ramps that have long delays and affect substantially the total travel time of the system.

Table 1. I-494 performance with existing and new control

	total travel time	ramp enter volume	aver speed
	veh-h	veh	mph
a.m. peak			
exst	1810	11130	56
new	1250	12000	56
p.m. peak			
exst	2620	16210	55
new	1820	19660	54

The resulting optimal thresholds were implemented during a field test in December 1992 - January 1993. The traffic performance in the test section, before and after the new control, was evaluated using the data collected from the loop detectors in the test section. Table 1 summarizes the traffic performance resulting from the new schedule along with that from the existing schedule computed for the same demand pattern. As the Table illustrates, the new control allows more cars to enter the freeway but reduces total travel time. For instance, in the 3-hour p.m. peak period, 21% more vehicles enter through the ramps, and the increased volume is accompanied by a 31% reduction in total travel time.

The substantial reduction in total travel time results from reduction in the waiting time at the ramps. By focusing on ramps with the longest waiting, the method alleviates traffic conditions at the worst performing ramps first, resulting in an overall travel time improvement.

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The new control method is being enhanced in a number of ways. The enhancements include incorporation of traffic predictors, direct interaction with intersections adjacent to ramp areas, and automation of the complete process so that it can be implemented on-line in an integrated hierarchical system that is truly adaptive.

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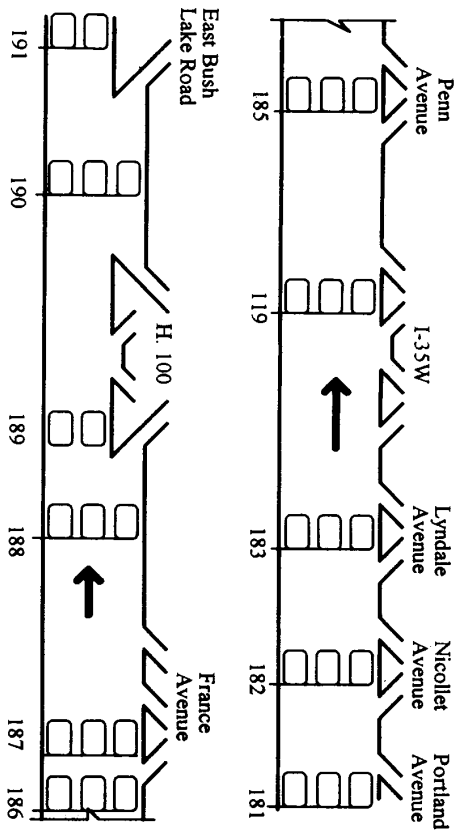

Figure 1. I-494 westbound test section

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